

Python Tools: UV

New Python Package Manager

uv replaces venv and pip

- **Speed Comparison:** uv is 10-100x faster than pip!
- Written in Rust
- uv replaces: venv, pip, pipx, and Python version management
- **Manages Python versions** automatically
- The tool that is used by AI MCP tools.

Installation

```
# Install uv
pip3 install uv

# UV examples
# Create virtual environment
uv venv

# Install packages (much faster!)
uv pip install pandas numpy requests
```

Depending on the system, the `pip` command may be used instead.

- Make sure `python` or `python3` is in your path.
- When the `pip/pip3` command is not available, try `python -m pip` or `python3 -m pip` instead.

Where the UV is Located

macOS/Linux:

```
~/.local/bin
```

Windows:

```
%APPDATA%\uv\bin
```

Be sure that `~/.local/bin` is in the path.

You can also use a symlink to copy the link to the path (not recommended, but useful sometimes).

```
# In the uv directory  
sudo ln -s ~/.local/bin/uv /usr/local/bin/uv
```

Problems That **UV** Solves

- Manage multiple Python versions — `uv python`
- Simplify virtual envs — `uv env`
- Avoid per-project package conflicts — `uv run`
- Do one-off, temporary installs (like `npx`) — `uvx`
- Globally install & manage CLI tools (like `npm -g`) — `uv tool install <pkg>` / `uv tool run <cmd>`
- Manage syncing environments - `uv sync`

Quick Examples

```
# Install and pick a Python version
```

```
uv python install 3.12
```

```
uv python pin 3.12
```

```
# Create & activate an env (auto-activation in shells supported)
```

```
uv env create
```

```
uv run python -V
```

```
# One-off run (no permanent install)
```

```
uv run ruff --version
```

```
# Install a global CLI tool
```

```
uv tool install ruff
```

```
uv tool run ruff check .
```

Manage multiple Python versions — **uv** **python**

UV automatically manages Python versions!

- **Downloads** Python versions as needed
- **Switches** between versions per project
- **No need** for pyenv, conda, or manual installs

💡 Think of it as: **Python version vending machine**

Installing Python Versions

```
# List available Python versions
uv python list

# Install specific Python version
uv python install 3.12
uv python install 3.11
uv python install 3.10

# List installed versions
uv python list --only-installed
```

UV downloads and manages Python for you!

Installed Directory

With `uv python`, Python is installed in the `~/share/uv/python` directory.

- With the `uv python find` command, we can find the exact location of the Python.

```
chos5@NKU023R7042 temp> uv python install 3.12
Installed Python 3.12.10 in 3.01s
+ cpython-3.12.10-macos-aarch64-none

> uv python find 3.12
~/local/share/uv/python/cpython-3.12.10-macos-aarch64-none/bin/python3.12
```

We can use this version of Python globally with a symbolic link.

```
# In the uv directory
sudo ln -s $(uv python find 3.12) /usr/local/bin/python312
```

Simplify virtual envs — `uv env`

- Creates isolated Python environments (like `python -m venv`)
- Same activation method: `source <env>/bin/activate`
- **Key advantage: 10–30× faster** than Python's built-in `venv`

```
# Speed comparison (approximate)
time python -m venv test_env      # ~2–3 seconds
time uv venv test_env             # ~0.1–0.2 seconds
```

Feature	uv venv	Python venv
Speed	⚡ Ultra-fast (Rust)	🐌 Slower (Python)
Python Discovery	Auto-finds Python versions	Requires explicit path
Built-in	No - External tool	Yes - Built into Python3
Dependencies	Requires uv installation	No extra installation

uv venv vs Python venv

uv venv (Smart Discovery)

```
# Automatically finds Python 3.11
# If Python 3.11 isn't in uv's cache,
# it's auto-installed before the venv is created.
uv venv --python 3.11 # .venv directory
my venv myenv          # myenv directory

# Works with python, python3, py, etc.
uv venv myenv --python 3.12
```

Python venv (Manual Path)

```
# Need exact Python executable
python3.11 -m venv myenv

# Or full path on some systems
/usr/bin/python3.11 -m venv myenv
```

Directory Structure: Almost Identical

`uv venv` creates `.venv/` directory, but we can specify any directory with `uv venv DIRECTORY`.

```
.venv/  
├── bin/           # Scripts (Linux/macOS)  
├── Scripts/      # Scripts (Windows)  
├── lib/          # Python packages  
├── include/      # Header files  
└── pyvenv.cfg    # Configuration
```

Both create the same structure with the same files!

Avoid per-project package conflicts — uv

run

Creating Projects

With UV, creating and using an isolated project with venv is straightforward.

```
# Method 1: Specify during project creation
uv init my-project --python 3.12
cd my-project
```

```
# Method 2: Set Python version for existing project
cd existing-project
echo "3.11" > .python-version
uv venv # Uses Python 3.11
```

```
# Method 3: Direct venv creation
uv venv --python 3.10
```

uv init command

It creates a project skeleton.

- Don't forget to make a venv: If you run `uv venv` in a directory with a `.python-version` (or in its parent), `uv` uses that Python version for the venv.

```
uv init my-project --python 3.12
cd my-project
uv venv
```

```
my-project/
├── .python-version      # Specifies Python version
├── pyproject.toml       # Project configuration & dependencies
├── README.md
└── .venv/               # Virtual environment (created when needed)
    ├── bin/
    ├── lib/
    └── pyvenv.cfg
```


Different Projects, Different Python

```
# Legacy project (needs older Python)
uv init legacy-app --python 3.9
cd legacy-app
uv add "django==3.2" # Old Django version

# Modern project (latest Python)
uv init modern-app --python 3.12
cd modern-app
uv add "fastapi" # Latest FastAPI

# Data science project (stable Python)
uv init data-project --python 3.11
cd data-project
uv add "pandas" "jupyter"
```

Python Version Priority (How UV Chooses)

1. **Command line:** `--python 3.12`
2. `.python-version` **file:** Project-specific
3. `pyproject.toml` : `requires-python = ">=3.11"`
4. **System default:** Whatever's available

Best Practice: Always specify in `.python-version`

uv add vs pip

Similarities

- Install Python packages from PyPI
- Support version specifiers (`requests==2.31.0`)
- Can install multiple packages at once
- Use installed packages in your project immediately

Differences

Feature	<code>uv add</code>	<code>pip</code>
Dependency Mgmt	Updates <code>pyproject.toml</code> & <code>uv.lock</code> automatically	No lock file by default
Speed	Much faster (parallel, prebuilt wheels)	Slower
Isolation	Works with <code>uv</code> venvs seamlessly	Requires manual <code>venv</code> handling
One-off Install	<code>uv run <pkg></code> without install	No direct equivalent
Cache	Shared global cache for re-use	Builds per venv

Example

```
# uv
uv add requests
uv remove requests          # Remove dependency

# pip
pip install requests
pip uninstall requests
```

The pyproject.toml is updated automatically.

```
[project]
name = "my-project"
version = "0.1.0"
description = "Add your description here"
readme = "README.md"
requires-python = ">=3.12"
dependencies = [
    "requests>=2.32.4",
]
```

Avoid per-project package conflicts — **uv** **run**

```
uv run python script.py
```

```
# uv add pytest
```

```
# uv add jupyter
```

```
uv run pytest tests/
```

```
uv run jupyter notebook
```

- **Purpose:** Run commands in **current project's environment**
- **Like:** Activated virtual environment
- **Think of it as:** "Run this in my current project"
- **Available:** Only within project directory

How `uv run python` Works

1. **Finds or creates** a project-specific virtual environment based on `pyproject.toml` and `uv.lock`.
2. **Temporarily activates** this isolated environment automatically.
3. **Runs your Python command or script inside** this managed environment with exact dependencies.
4. **Ends the session** without manual activation or cleanup—the environment persists for reuse.

Key benefit: Seamlessly manages environment syncing and activation on-the-fly, eliminating manual steps and avoiding dependency conflicts.

- **Purpose:** Run commands in **current project's environment**
- **Like:** Activated virtual environment
- **Think of it as:** "Run this in my current project"
- **Available:** Only within project directory

1. Create a project and add a dependency

```
uv init my-project  
cd my-project  
uv add requests
```

2. Create a script that uses the dependency

```
echo 'import requests; print("Works!")' > script.py
```

3. Compare the two approaches:

```
python script.py          # Might fail (no requests available)  
> ModuleNotFoundError: No module named 'requests'  
  
uv run python script.py    # Works (uses project environment)  
# PAHT/my-project/.venv/bin/python script.py  
> Works!
```

Do one-off, temporary installs (like `npx`) — `uvx`

Think of `uvx` like Node.js's `npx` but for Python tools!

- **Run Python tools** without permanent installation
- **Isolated execution** - no conflicts with your system
- **Automatic cleanup** - tools are cached but don't clutter your environment
- **Fast execution** - reuses cached installations



Perfect for: Testing tools, one-time scripts, CI/CD pipelines

uvx vs Traditional Installation

Traditional Way (Permanent Install)

```
# Install tool globally (clutters system)
pip install black
black my_file.py

# Or in a virtual environment (manual setup)
python -m venv temp_env
source temp_env/bin/activate
pip install black
black my_file.py
deactivate
rm -rf temp_env # Manual cleanup
```

With uvx (Temporary)

```
# One command – automatic isolation & cleanup
uvx black my_file.py
```

How **uvx** Works Behind the Scenes

1. **Check cache:** Is the tool already downloaded?
2. **Create isolated environment:** Temporary virtual environment
3. **Install tool:** Only if not cached
4. **Execute command:** Run the tool with your arguments
5. **Keep cache:** Tool stays cached for future use
6. **Clean isolation:** Environment is temporary

Usage Examples

1. Format a Python file

```
echo "x=1;y=2;print(x+y)" > messy.py
```

```
uvx black messy.py
```

```
cat messy.py
```

2. Create a simple web server

```
echo "<h1>Hello World</h1>" > index.html
```

```
uvx http.server 8000
```

```
# Visit http://localhost:8000
```

3. Analyze a requirements file

```
echo "requests==2.25.0\ndjango==3.0" > requirements.txt
```

```
uvx safety check -r requirements.txt
```

Common **uvx** Patterns

Testing Different Versions

```
# Test with different Black versions
uvx black==22.0.0 my_file.py
uvx black==23.0.0 my_file.py
uvx black==24.0.0 my_file.py
```

One-off Scripts

```
# Run a script that needs specific packages
uvx --with pandas --with matplotlib python analyze_data.py
```

Quick Utilities

```
# Quick file serving
uvx http.server 3000
# Quick code formatting
uvx autopep8 --in-place *.py
```

Useful UVX Tools

Code Formatting & Linting

```
# Format Python code with Black
uvx black .
# Lint code with Ruff
uvx ruff check .
# Sort imports with isort
uvx isort .
```

Web Development

```
# Create a Django project without installing Django
uvx django-admin startproject mysite
# Run HTTP server for static files
uvx http.server 8000
```

Package Management Tools

```
# Check package vulnerabilities
uvx safety check
# Analyze dependencies
uvx pipdeptree
# Check outdated packages
uvx pip-review
```

Data Science & Analysis

```
# Quick data analysis
uvx pandas-profiling data.csv
# Convert notebooks
uvx nbconvert notebook.ipynb --to html
# Run Jupyter lab temporarily
uvx jupyterlab
```


uvx vs npx Comparison

Feature	uvx (Python)	npx (Node.js)
Purpose	Run Python packages	Run npm packages
Installation	Temporary execution	Temporary execution
Caching	Yes (reuses cached)	Yes (reuses cached)
Isolation	Automatic venv	Node modules
Cleanup	Automatic	Automatic

Globally install & manage CLI tools (like `npm -g`) — `uv tool install <pkg>` / `uv tool run <cmd>`

Don't install everything globally - be selective

- Use `uv tool` for tools you'll use repeatedly
 - Use `uv tool run` for experimentation
- Use `uv run` for project-specific commands

uv tool (Persistent Tools)

```
uv tool install black  
uv tool uninstall black
```

- **Purpose:** Install tools **permanently** in isolated environments
- **Like:** `pipx` - each tool gets its own space
- **Think of it as:** "Add this to my workshop toolbox"
- **Available:** Everywhere, across all projects

How UV Organizes Tools

Tools are installed in `~/.local/share/uv/tools`.

```
~/.local/share/uv/tools/  
├── black/                                # Each tool gets its directory  
│   ├── bin/                             # Executable  
│   │   └── black  
│   ├── lib/  
│   │   └── python3.11/  
│   │       └── site-packages/  
│   └── pyvenv.cfg  
├── cookiecutter/  
│   ├── bin/  
│   ├── lib/  
│   └── pyvenv.cfg
```

*notice that `~/.local/bin/black` is linked to
`~/.local/share/uv/tools/black`*

~/.local/bin/black (Direct execution) vs uv tool run black (UV-managed execution)

They are the same in most cases: functionally identical: The `~/.local/bin/black` is a **wrapper script created by UV** that does essentially the same thing as `uv tool run black`. Both:

- Use the same isolated environment
- Have the same dependencies
- Run the same version of Black

When They Might Differ

1. Environment Resolution

```
# UV tool run: Always uses UV's managed environment  
uv tool run black my_file.py
```

```
# Direct call: Uses the wrapper, but could theoretically be overridden  
~/.local/bin/black my_file.py
```

2. Path Issues

```
# If ~/.local/bin is not in PATH:  
~/.local/bin/black my_file.py # Works (full path)  
black my_file.py              # Might not work  
uv tool run black my_file.py  # Always works
```

3. Tool Management

```
# UV knows about the tool
```

```
uv tool list
```

```
uv tool upgrade black
```

```
# Shows black
```

```
# Updates black
```

```
# Direct execution doesn't involve UV's management layer
```

When to Use `uvx` vs `uv tool install`

Use `uvx` for:

- **One-time tasks:** Formatting, testing, analysis
- **Trying tools:** Before deciding to install permanently
- **CI/CD scripts:** Temporary tool usage
- **Different tool versions:** Testing multiple versions

Use `uv tool install` for:

- **Daily tools:** Tools you use regularly
- **CLI utilities:** Commands you want in PATH
- **Development workflow:** Part of your regular setup

Manage syncing environments - **uv sync**

(Old way) Using requirements.txt and freeze

```
# Using uv (fast & modern)
uv add -r requirements.txt
uv pip freeze > requirements.txt
```

```
# Using pip (classic way)
pip install -r requirements.txt
pip freeze > requirements.txt
```

(New Way) UV Lock Files Like package-lock.json

```
# Generate lock file (exact versions)
uv lock

# Install from lock file (reproducible builds)
uv sync

# Update dependencies
uv lock --upgrade
```

uv.lock ensures everyone gets **exact same versions**

uv lock and pyproject.toml

File	Purpose	Who Should Have It	Action	Edit?
pyproject.toml	Declare broad project dependencies	Everyone	Add/remove dependencies	Yes
uv.lock	Lock exact dependency versions and environment	Everyone (shared)	Ensure reproducible installs	No (auto-managed)

Work Process Example

1. Develop and specify dependencies in `pyproject.toml`.
2. Run commands `uv add <package>` or manually edit `pyproject.toml`.
3. Run `uv sync` or `uv lock` to update `uv.lock` with resolved versions.
4. Commit both files (`pyproject.toml` and `uv.lock`) to your version control.
5. Other developers or CI systems run `uv sync` to install the exact locked environment.
6. Avoid using `pip install -r` when using uv with lockfiles; use `uv sync` instead.